



Polynomial regression using scikit-learn

I am trying to use scikit-learn for polynomial regression. From what I read polynomial regression is a special case of linear regression. I was hoping that maybe one of scikit's generalized linear models can be parameterised to fit higher order polynomials but I see no option for doing that.

I did manage to use a Support Vector Regressor with a poly kernel. That worked well with a subset of my data, but it takes much too long to fit larger data sets so I still need to find something faster (even if trading some precision).

Am I missing something obvious here?

regression machine-learning large-data polynomial scikit-learn

asked May 11 '13 at 20:00



Mihai Damian

173 1 1 6

3 Answers

Given data $\mathbf{x}$, a column vector, and $\mathbf{y}$, the target vector, you can perform polynomial regression by appending polynomials of $\mathbf{x}$. For example, consider if

$$\mathbf{x} = \begin{bmatrix} 2 \\ -1 \\ \frac{1}{3} \end{bmatrix}$$

Using just this vector in linear regression implies the model:

$$y = \alpha_1 x$$

We can add columns that are powers of the vector above, which represent adding polynomials to the regression. Below we show this for polynomials up to power 3:

$$X = \begin{bmatrix} 2 & 4 & 8 \\ -1 & 1 & -1 \\ \frac{1}{3} & \frac{1}{3^2} & \frac{1}{3^3} \end{bmatrix}$$

This is our new data matrix that we use in sklearn's linear regression, and it represents the model:

$$y = \alpha_1 x + \alpha_2 x^2 + \alpha_3 x^3$$

Note that I did not add a constant vector of 1's, as sklearn will automatically include this.

edited May 11 '13 at 22:10

answered May 11 '13 at 22:02



Cam.Davidson.Pilon

5,210 1 22 44

Ok, this makes sense. Thanks for a great answer. – Mihai Damian May 12 '13 at 8:52

Theory

Polynomial regression is a special case of linear regression. With the main idea of how do you select your features. Looking at the multivariate regression with 2 variables: x_1 and x_2 . Linear regression will look like this: $y = a_1 * x_1 + a_2 * x_2$.

Now you want to have a polynomial regression (let's make 2 degree polynomial). We will create a few additional features: $x_1 * x_2$, x_1^2 and x_2^2 . So we will get your 'linear regression':

$$y = a_1 * x_1 + a_2 * x_2 + a_3 * x_1 * x_2 + a_4 * x_1^2 + a_5 * x_2^2$$

This nicely shows an important concept **curse of dimensionality**, because the number of new features grows much faster than linearly with the growth of degree of polynomial. You can take a look [about this concept here](#).

Practice with scikit-learn

You do not need to do all this in scikit. Polynomial regression is already available there (in 0.15 version. Check [how to update it here](#)).

```
from sklearn.preprocessing import PolynomialFeatures
from sklearn import linear_model

X = [[0.44, 0.68], [0.99, 0.23]]
vector = [109.85, 155.72]
predict= [0.49, 0.18]

poly = PolynomialFeatures(degree=2)
X_ = poly.fit_transform(X)
predict_ = poly.fit_transform(predict)

clf = linear_model.LinearRegression()
clf.fit(X_, vector)
print clf.predict(predict_)
```

edited May 23 at 12:39

 Community ♦
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answered Nov 19 '14 at 11:32

 Salvador Dali
528 6 16

- What if I do not want to have an interaction terms as $x_1 \cdot x_2$, do i have to construct $X_$ manually? there's a parameter "interaction_only" in PolynomialFeatures() constructor, and by default it is False. But setting it to True makes the opposite of what i want: it keeps ONLY the interaction terms, and do not keep x_1^2 , x_2^2 , etc. – DenisFLASH Dec 1 '14 at 14:59
- The link to youtube claims that the video does not exist anymore. Do you have any other link to that? – Markon May 10 '16 at 9:06
- @Markon any video from this list is good enough: [youtube.com/results?search_query=curse+of+dimensionality](https://www.youtube.com/results?search_query=curse+of+dimensionality) – Salvador Dali May 10 '16 at 16:01
- @SalvadorDali that is what dimensionality reduction is for – user3916597 Jul 6 '16 at 4:49
- I wonder if we should center the data before or after applying PolynomialFeatures ? – renakre May 5 at 20:09

In case you are using a multivariate regression and not just a univariate regression, do not forget the cross terms. For instance if you have two variables x_1 and x_2 , and you want polynomials up to power 2, you should use $y = a_1x_1 + a_2x_2 + a_3x_1^2 + a_4x_2^2 + a_5x_1x_2$ where the last term ($a_5x_1x_2$) is the one I am talking about.

edited Apr 4 '14 at 10:25

 Nick Stauner
8,970 5 27 57

answered Apr 4 '14 at 9:36

 Vermeer Grange
21 2